Wireless Technology

Tasks

1. Scan for available Wi-Fi networks on your laptop or phone and list the SSIDs you find, noting which ones are open (unsecured) and which require a password.

Ans.:-

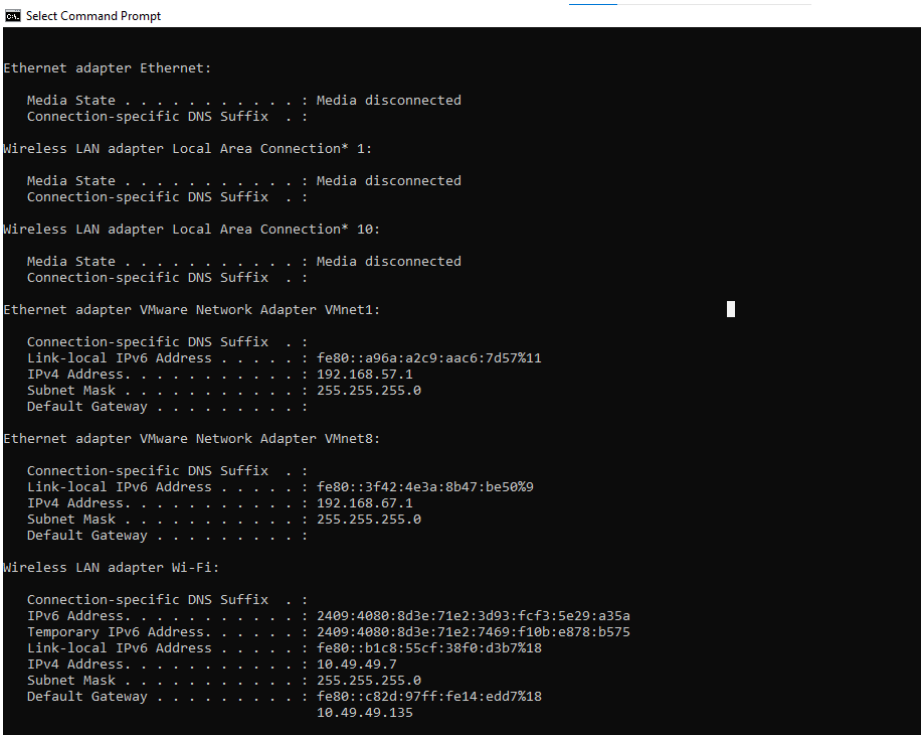
Sr. No.	SSID (Wi-Fi Name)	Security	Password Required?
1	Bajrang 4	WPA2/WPA3	Yes
2	VIVO y22	WPA2/WPA3	Yes

**2. Set up a mobile hotspot on your phone, connect a laptop to it, and check what IP address your laptop receives from the hotspot.

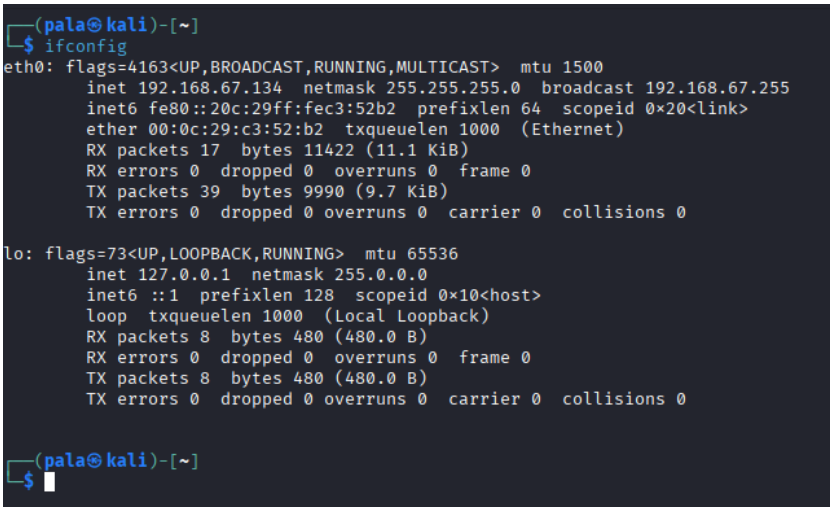
Hint: On Windows, use 'ipconfig' in Command Prompt; on Mac/Linux, use 'ifconfig' in Terminal.**

Ans.:-

Windows :-



KALI Linux :-



3. Research and explain in your own words what WPA2 and WPA3 are, and list two differences between them relevant to security.

Ans.:-

WPA2 and WPA3 :-

WPA2 (Wi-Fi Protected Access 2) and WPA3 (Wi-Fi Protected Access 3) are security standards used to protect Wi-Fi networks from unauthorized access and attacks.

POINT	WPA2	WPA3
SECURITY METHOD	Uses AES-CCMP encryption	Uses stronger AES-based encryption and improved security mechanisms
PASSWORD PROTECTION	More vulnerable to offline password-guessing attacks if the password is weak	Uses SAE (Simultaneous Authentication of Equals), which provides better protection against offline password-guessing
SECURITY LEVEL	Strong when configured with a strong password	Generally provides stronger protection, especially against password-based attacks

Two Important Security Differences

1. Password protection: WPA3 uses SAE, which makes offline password-guessing attacks significantly harder than with WPA2-PSK.
2. Stronger security: WPA3 provides improved protections and security requirements compared with WPA2, particularly for modern Wi-Fi networks.

4. Find a news article or real-world case from the past 3 years where a public Wi-Fi network was used in a cyber attack, and summarize how the attack happened and what could have prevented it.

Ans.:-

Real-World Case: Fake Public Wi-Fi Attack in Australia

A good example occurred in 2024 in Australia. Australian Federal Police alleged that a man created fake “free Wi-Fi” networks at Perth, Melbourne and Adelaide airports and on domestic flights. The fake networks copied legitimate Wi-Fi names, an attack commonly called an “evil twin” attack.

How the attack happened :

1. The attacker created Wi-Fi networks that looked like legitimate airport or airline Wi-Fi.
2. People connected to the fake networks, believing they were official.
3. Users were redirected to fake login pages asking for email or social-media credentials.
4. The entered information was allegedly saved by the attacker and could potentially be used to access other personal information.

What could have prevented it:

- Verify the exact Wi-Fi name (SSID) with airport/airline staff before connecting.
- Avoid entering email, social-media, banking, or other sensitive credentials into unexpected Wi-Fi login pages.
- Use a VPN when using public Wi-Fi.
- Disable unnecessary file sharing on the laptop/phone.
- Use HTTPS websites and enable multi-factor authentication on important accounts.

Public Wi-Fi can be risky because attackers can create fake networks that look legitimate. Users should verify the network and avoid entering sensitive information on suspicious captive portals.